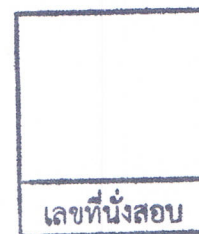


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 2

Year: 2014

Subject: 392152 Chemistry II

Section: 15-16

Date: 12 March 2015

Time: 10.00-12.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Only calculators are allowed in the examination.
4. This exam is a closed book examination.
5. Electronic communication devices are not allowed in the examination room.
6. The examination has 6 pages (including this page), 3 sections and a total score of 60 points.
7. Write all your solution and answer on this examination sheet.

Relative atomic mass of

H = 1 He=4 C= 12 N= 14 O=16 Na=23 S=32 Cl=35.5

Gas constant; $R = 0.082 \frac{\text{atm.L}}{\text{mol.K}}$ or $8.314 \frac{\text{J}}{\text{mol.K}}$

Part 1. Multiple Choice (11 points)

Choose the letter of the choice in the answer table that would be the best completes the statement or answers the question.

Question no.	a	b	c	d
1				
2				
3				
4				
5				
6				

Question no.	a	b	c	d
7				
8				
9				
10				
11				

- A change in solid to liquid is called _____.
a. sublimation b. melting c. evaporation d. deposition
- When vaporization occurs at the surface of a liquid that is not boiling, the process is called?
a. melting b. freezing c. evaporation d. sublimation
- The ideal gas equation is given by $PV = nRT$. Which one of the following conditions is true according to Avogadro's hypothesis?
a. $P \propto 1/V$ ($T = \text{constant}$) b. $V \propto T$ ($P = \text{constant}$)
c. $V \propto n$ ($P, T = \text{constant}$) d. $P \propto T$ ($n = \text{constant}$)
- What law may be stated qualitatively as "When you squeeze a gas, it gets smaller"?
a. Boyle's law b. Charles' law c. Ideal gas law d. Graham's Law
- The vapor pressure of a substance increase as _____ increase
a. viscosity b. molecular mass c. temperature d. intermolecular force
- Sedimentation occurs easily in _____.
a. solution b. suspension c. colloid d. compound
- Molality is defined as _____.
a. kg solute/kg solution b. moles solute/ kg solvent
c. moles solute/ liter solution d. moles solute/mole solvents
- Which of the following presents the highest pressure?
a. 760 torr b. 1×10^5 Pa c. 2 atm d. 152 mm Hg

9. Under sea level, water boils at a _____ temperature due to the _____ atmospheric pressure.
- higher, increase
 - lower, increase
 - higher, decrease
 - lower, decrease
10. The normal boiling point of chloroform, which has a higher vapor pressure than water at 100°C, is _____.
- higher than the normal boiling point of water
 - lower than the normal boiling point of water
 - the same as the normal boiling point of water
 - unable to be measured
11. In which condition that the behavior of a real gas is similar to the behavior of an ideal gas?
- high temperature and high pressure
 - high temperature and low pressure
 - low temperature and high pressure
 - low temperature and low pressure

Part 2 Filling in the blanks of the following statements with one of these words: (14 points)

plasma gas faster sublimation liquid slower dissolve vibrate

- Sugar dissolves in hot water than it does in cold water.
- Iodine, a piece of grey solid, changes directly into a gas when heated gently.
This is called
- In solids, the particles
- The spacing of particles in a is greater than in a
- Heavy gases move than light ones.
- is not a common state of matter under ambient conditions.

Part 3 Answer (35 points)

- A sample of helium occupies a volume of 160 cm³ at 100 kPa .What volume will it occupy if the pressure is adjusted to 80 kPa and if the temperature remains unchanged? (2 points)

2. At a temperature of 298 K, a certain amount of CO_2 gas occupies a volume of 6 L. What volume will the gas occupy if its temperature is reduced to 273 K? (2 points)

3. A sample of gas occupies 12.0 L at 240°C under the pressure of 80.0 kPa. At what temperature would the gas occupy 15.0 L if the pressure increased to 107 kPa? (3 points)

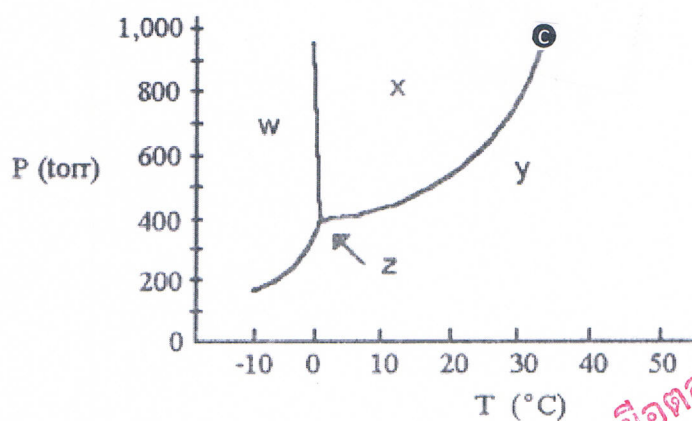
4. Calculate the pressure of a 1.2-mol sample of methane gas in a 3.3 L container at 25°C .

(2 points)

5. Calculate the molecular mass of a gas if a 9.21-g sample occupies 4.30 L at 127°C and a pressure of 342 torr.

(3 points)

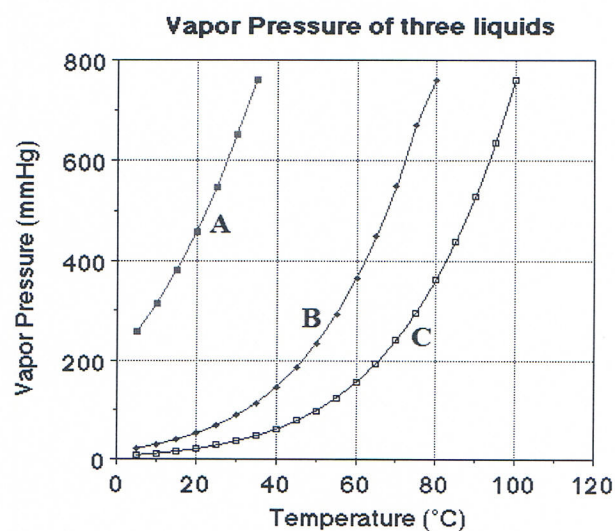
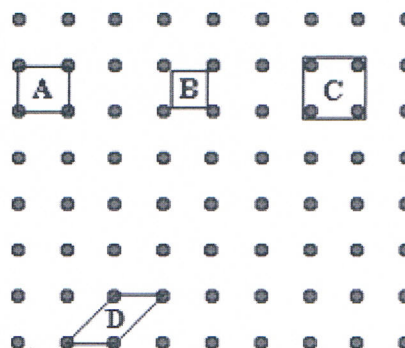
6. Oxygen gas and sulfur dioxide (SO_2) gas are at the same temperature. What is the ratio of the effusion rates of their molecules? (3 points)



7. According to the phase diagram on the left: (3 points)

- The critical point is _____.
- The tripple point is _____.
- The phase of w is _____.

8. Which parallelograms in the figure on the right are unit cells? (3 points)



9. According to the plot of vapor pressure versus temperature on the left, determine the normal boiling point of:

(3 points)

Liquid A : _____

Liquid B : _____

Liquid C : _____

10. A solution is prepared by dissolving 5 g of NaCl in 245 g of water to make 250 mL of solution.

Finding

(8 points)

- a). percentage concentration by mass (%w/w)
- b). percentage concentration by mass per volume (%w/V)
- c). the molality of solution
- d). the mole of NaCl

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11. Calculate the mass of HNO_3 in 500 mL of 2.0 M HNO_3 solution

(3 points)